DE FIELD GUIDE FIELD GUI

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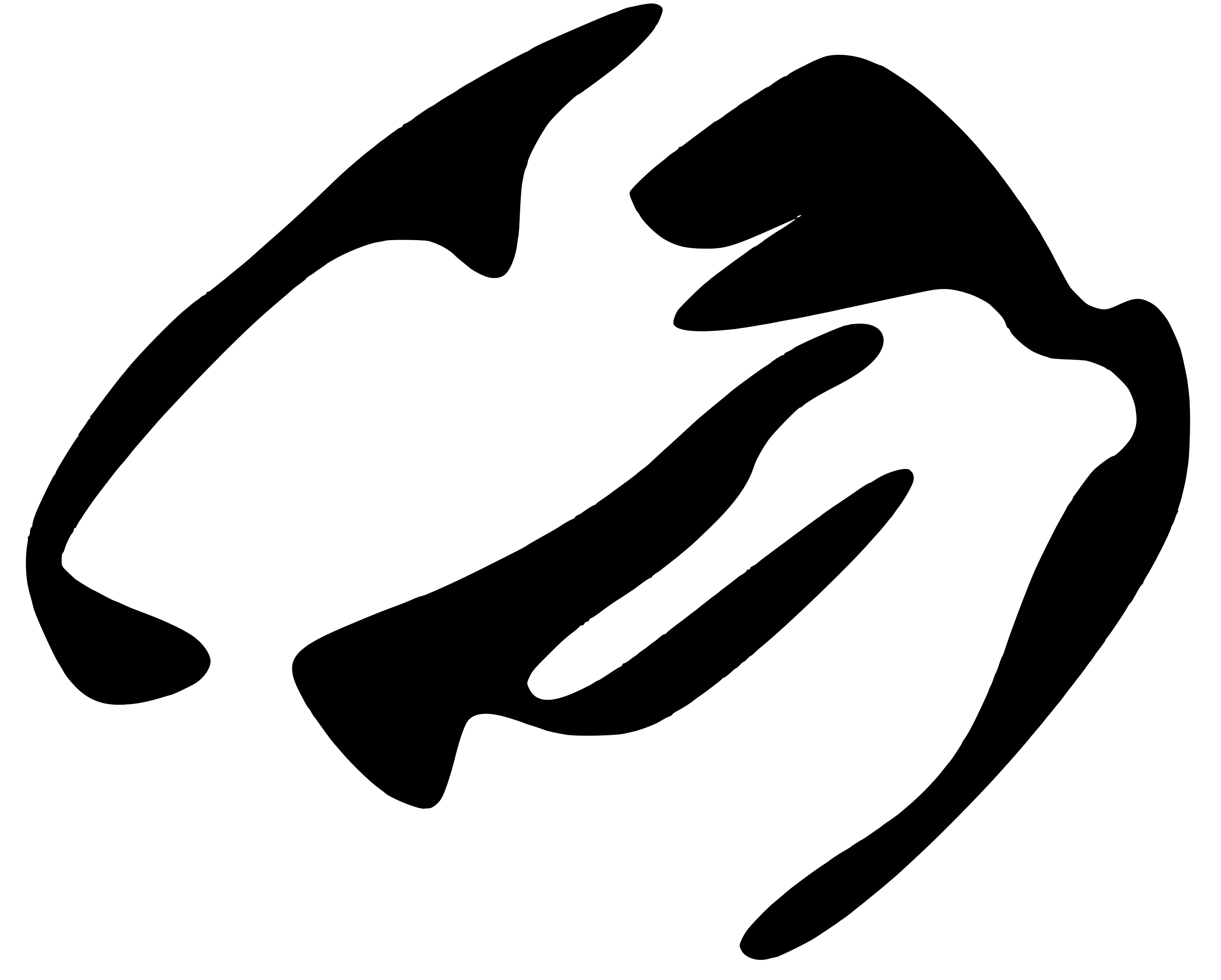
time travel cv playground

2025

SUPRA

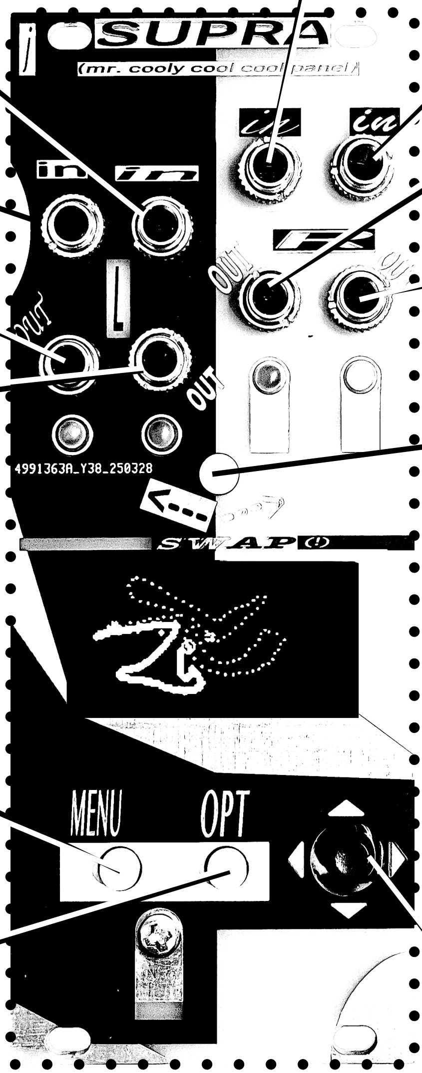



Bounce

OVERVIEW

Supra is a multimode module with 4 modes. Each mode is an experiment in 'supra-time' a term coined by Curtis Roads in 'microsound' to describe changes that happen at the scale beyond one musical performance or event. Supra is a set of tools that slowly change on their own very slow timeline. The modes are called: Cryptex, Abacus, Stamp, and Chime.

Supra is divided into 2 "hands" Left and Right. Each hand has 2 gate inputs and 2 gate/cv outputs. Any of the 4 modes can be placed into either hand (including 2 copies of the same mode, which will operate independently). Pressing swap button chooses which hand you're 'looking at' meaning which is displayed on the screen, and controlled by the joystick, menu, and opt buttons. The hand you're not looking at will continue to run in the background. Holding the swap button lets you place a new mode in the current hand.



Left Hand in 2

Right Hand in 1

Right Hand in 2

Left Hand in 1

Right Hand out 1

Right Hand out 2

Left Hand out 1

Left Hand out 2

SWAP BUTTON:
SWITCHES WHICH HAND YOU'RE VIEWING AND CONTROLLING. THE OTHER HAND CONTINUES TO RUN IN THE BACKGROUND. **HOLD** TO PLACE NEW OBJECT IN CURRENT HAND

MENU WITH EXTRA FEATURES OF CURRENT MODE

OPT EXTRA BUTTON FOR USE IN SOME MODES

JOYSTICK:
UP DOWN LEFT RIGHT AND **CLICK** (PUSH IN)

SUPRA

cryptex

-.-..-.-.--.--.-.-..-

In 1: clock
in 2: reset
out 1: sequence gates
out 2: in split mode outputs dash gates (out1 is dot gates)

Morse code translation rhythm scavenger

Cryptex is an encoded epoch scale reading of Homer's Odyssey. Every 3 hours, 3 new words from the Odyssey are displayed (in order), alongside their morse code translations. The user can then select portions of the translation to loop through as dot-dash based gate rhythms, and rotate the entire translated sequence to change which part plays in the loop

Controls:

- Joystick ◀ ▶ rotates the entire sequence, changing what is inside of the looped section

- Joystick *click* switches to edit mode, use ◀▶▼▲ to move the loop point around, changing the sequence length

Dots

E .
I . .
S . . .
H …. .

Dashes

T -
M --
O ---

Just C

C -.-.

Dot Dashes

A .-
W .--
J .---

Dash Dots

N -.
D -..
B -...

Dash Sandwiches

R .-.
P .--.

Dot Sandwiches

K - . -
X - . . -

Double Dashes

G --.
Z --..

Double Dots(um or 3)

U ..-
V ...-

Jumping Dashes

L .-..
F ..-.

Crouching Dots

Y -.--
Q --.-

STAMP

Graphical Rail Riding CV Generator

In 1: reset loop (see menu reset modes) | In 2: n/a | out 1: CV value | out 2: n/a

Stamp generates CV by tracing a portion of an image in a loop. As the outline of the image is traced, the closer the tracing point is to the center of the screen, the higher the voltage. Also, the closer the tracing point to the center, the faster it moves (this attraction forced is called gravity). Stamp contains 365 drawn images, a different one shown for each day of the year.

Controls: (in relation to the '*head*' of the loop which looks like this:

▶ right extends the length of the loop, moving away from the head
◀ left shortens the length of the loop, moving towards the head

▲ up shifts the position of the loop along the image, towards the head
▼ down shifts the position of the loop along the image, away from the head

The images in STAMP have a different theme each month.

make notes here on what you think they are

JAN	JULY
FEB	AUG
MAR	SEP
APR	OCT
MAY	NOV
JUN	DEC

Abacus

Poly-Constraint Rhythm Sequencer

In 1: clock   in 2: reset   out 1: sequence 1 gate   out 2: sequence 2

Abacus Is a Step Sequencer. Each day you get 2 new blank sequences of a random length to place steps in. You also get one special 'skip-step' type per day to place in your sequences.

Use joystick ◀ ▶ to navigate inside of a sequence and click to add, or remove the current step. Use ▲ ▼ to navigate between the two sequences. Press OPT to switch between placing normal steps and placing your skip-step (shown in top left) which will only play every 2nd, 3rd, etc. time its passed (depending on the type)

Time for some math! With 2 sequences of set length, simply multiply their lengths to determine how many input pulses it takes to have them sync back up on step one. So for lengths 8, and 5 - every 40 input pulses they will be synced on step 1. Skip steps of course complicate this and allow for combinatorially complex rhythms.

Use the circle grid to work out Abacus math and simulate rhythms

CHIME SCHEDULIZED MELODIC SEQUENCER

In 1: clock   In 2: reset   Out 1: sequence cv (v/oct pitch)   Out 2: n/a

Chime lets you sequence notes within a provided scale. The scale is scheduled: each day of the week has 2 scales, one for before noon, and one for after noon. So, every Friday morning will have the same scale, and every Wednesday evening, etc. When a season changes, ie, on the first day of spring, all of the scales in the schedule are swapped out with the scales selected for that season. So there are 14 weekly scheduled scales, and 4 banks of scales, one for each season (56 total scales). Use joystick ◀▶▲▼ to navigate the top library section (the top section is upper octave) click to place a note in the sequence. Pressing opt switches to 'edit' mode in which you can navigate within the sequence, and click to 'pick up' a note (which will start flashing) and move it around, clicking again to place the note back down

Here is a chart of scales expressed in number format (day and night in that order) according to Ian Ring's scale number format. To find out more about the scale you can go to:

ianring.com/musictheory/scales/num

For example:
ianring.com/musictheory/scales/351

SUMMER

SU	2741	and	2731
MO	1709	and	1967
TU	1451	and	1459
WED	2773	and	1493
Th	1717	and	1845
FR	1453	and	2475
SA	1387	and	1395

FALL

SU	1009	and	505
MO	351	and	501
TU	637	and	3877
WE	1119	and	3351
TH	3271	and	3683
FR	1777	and	1931
SA	3467	and	3725

WINTER

SU	1507	and	1423
MO	1509	and	981
TU	701	and	3403
WE	1815	and	1863
TH	2619	and	2955
FR	2845	and	1595
SA	695	and	2899

SPRING

SU	427	and	2261
MO	1333	and	1429
TU	373	and	2603
WE	2265	and	2253
TH	469	and	3157
FR	1365	and	2457
SA	723	and	813

AMBITIONS

time is weird - I had bigger ambitions for supra but the release is a month out when I'm writing this and I've got to put something out - deadlines keep me alive, without them there's just infinite sludge, goopy time that can quick sand you. I had set the release date a year ago in my mind - it's been a year 1/2 development - some months are 95% thumb twiddling, some weeks produce months of work. Time in projects has a massively fluctuating exchange rate - an hour isn't worth an hour.

OBJECTS

I don't really 'believe' in Objects - they're not a truly important part of life or at least not what I want to dedicate my time to. Especially creating new objects seems almost perverse, let alone new electronics as we shovel old ones away. But here we are, I've found myself as the objet maker and you an object consumer. What I love is making music, and playing with thing - games and systems. I've found myself at times craving to create something that beckons play - that induces that feeling - making the world in which the music is made, not the music itself. Making something for the people that make. I felt that as someone *who has spent a lot of time in different making worlds, that I could say something interesting with one of my own. I could inform a way of doing. Digital objects are amazing and I use them all the time, but there's something disposable about them. Something inherently practical tasting about them. Computer worlds for making bias towards producing an outcome - not getting lost in the messing-aroundery, which is maybe a synthesizer's most valuable offering. Physical making objects have controls, have form, have harder decisions so more constraints - are rarer, more valuable by way of meat space attention. And Systems - Systems have no timeline of bars, no official starts and ends, no render, or fast forward, or undo - an impracticality that makes them feel closer. So, for the world I wanted to express, I needed something physical, and systems based. And thus I arrived at Eurorack - as imperfect as it is, one of the few places which allowed playing in a physical system.*

DEVELOPMENT LOG

These are some little notes I made thinking back on each object in Supra, and how it was developed, what my life was like, the challenges etc.

CRYPTEX - Dec 2023

Cryptex was the first mode for supra, so its hard to remember when what started. I had a back of the brain interest in morse code - a friend made a morse code game about sinking ships by tapping out codes on a real key - i made a little pico-8 morse code sketch - i used morse code in a piece of music for work as a way to hide a secret from the developer. I liked the endless combination of rhythms made up entirely of the stilted pulses 1 or 3 counts long, which feels so different than how we usually think of rhythms. And i liked the idea of sifting through a pile of stuff, finding your own rhythms in the heap - creating through curating. I started working on cryptex while moving around a lot - in the Netherlands, Greece, New York - makes it feel like I've been working on it for so many years. The visual style i owe entirely to AC - I was showing him the mode at the time, really frustrated with some technical bug, and he just looked at it for one minute and then scribbled out a little diagram and said “maybe it could look like this” - i was really annoyed at the time because i didn’t want to deal with polish or visuals until later, but i ended up basically just recreating his sketch, and that visual style directs the way you engage with the mode beyond aesthetics. What the text should be was a big question mark for a really long time - at the very end of development i started to look for non copyrighted works that i could use, and the odyssey seemed so perfectly symbolic of length as a concept (and it has a lot of interesting/weird words). Im pretty happy with how cryptex turned out - theres something rhythmically significant about the way morse code sounds. Its the only class of rhythm that i can think of that might be so uniquely recognizable by non musicians - it has a cadence that we know, if more by feel than technical knowledge.

ABACUS - 2024 (late winter, early spring)

It's hard to remember when I actually started abacus - I remember working on it alongside AC and another friend, all doing wildly different projects - we all had desks in our common space, a super long thin apartment. I would sit and play with my prototype, going back and forth tweaking things about it. There's something magical about a rhythm in 2 parts, it's just perfect, something is very natural feeling about it. Abacus is a sort of successor to my first module Pet Rock, which generated a dual channel rhythm per day. I wanted it to serve as a marker of what I was trying to do with Supra - generating a tool to play with, rather than generating the content itself. Years ago I had sat with a little Swedish synth not to be named, and played with a sequencer mode with 2 oddly lengthed sequences, which I would try to randomly set, and then create patterns within. Abacus is basically an attempt to recreate that experience. I wanted something with simple design, where the fun came from you.

STAMP - summer 2024

Spring 2024 was rough - I was really sad about life things, and trying to make progress on the supra but doing more sitty aroundey tweakey work than anything real. In July I took a big pacific northwest trip visiting friends along the coast in Vancouver, Seattle and Portland. The trip made me feel better about a lot of things. I suddenly felt less stagnant, more curious again. I was sort of just absorbed in the environment, especially in vancouver - a lot of nice emptier time there, wandering long, and looking a things in the city - I went and camped on the tiny Keat's Island, off of the sunshine coast (shout out to the ferry white spot). I took the amtrak down to Seattle, right on the ocean, with a huge golden sunset, edge to edge in the big spread windows. That is to say, that I arrived in Seattle refreshed, if not some more dramatic word. My brain was ready to think upside down again. I stayed there at a friend's house studio, and accompanied him to his office one day which was kind of a bungalow of friends. There everyone was doing actual work, so I tinkered away on supra. I had been dipping my toe into bitmap images and pulled up the code. I wanted to start working on something that used cv not just gates, and I started playing there with how an Image's edge could be detected and traced if it was drawn correctly by hand. I drew a test image and by the end of the afternoon I had the visual framework and gravity movement of stamp working. It was such pure fun, translating something I had drawn into the world of the machine. Through the course of development represented more technical work and issues than anything else in the module, and the technical parts went through a lot of painful iterations - but I think it's the most *fun* mode - it's the best toy, and that's important, and makes me happy.

CHIME - summer 2024

A lot of the time after chime was sort of a hail mary mode. I wanted to have another mode that uses cv, and it took sooooo much longer to get stamp working than I expected. Then a lot of the time after that was spent designing and iterating the hardware - everything up to that point was made on a little piece of prototype board with off the shelf stuff. I had been away from programming as my main task for a bit. I knew that 4 was the absolute minimum for a multimode module in my head, so I started frantically working on chime last minute. The idea for time of day effecting scales was in the back of my mind for a long time, based on the concept from indian Raga. I played around with the scale being a random collection of notes, but that just didn't sound very good. I found this amazing site that compiles every conceivable scale and that set off the idea that the scales should be authored/picked. Chime has been the most thrown together mode for sure - everything has been on a too tight timeline, but I think it plays quite nicely with the rest of Supra. It makes the module a lot more versatile and provides a lot of combinatorial opportunities. I worked on chime alongside my friend Karp, mostly silently and frustratedly. It being the last mode also made me a little sad because at some point i envisioned more like 8 modes for supra, but alas, the module had to come out. Thinking about chime makes me feel stressed out hehe

The end of Supra development has been sort of a shit show. I left things too late, and have been scrambling to finish everything - I hate making decisions, so a lot of end of the line decisions had to be made, and then implemented. Oops! Guess it always feels like that to some degree, but it's been a wild ride - I'm not even finished as I write the manual. Here's hoping for the best :)

More Ramblings

What to do about cost? I released my first module 'Pet Rock' at cost of production - and Supra is sold at about cost of production (I've learned it's very difficult to reach perfect equilibrium - but no money goes back to me, anything more goes into more modules and development, anything less gets taken from that pool). I have the luxury of a fancy (and very fun) composition job which means I live comfortably completely outside of Eurorack - most makers don't have that. A lot of people misunderstand that I think everyone should sell at cost. That feels unfair to expect. What I do expect is for makers to acknowledge that Eurorack inaccessible and find a way to help change that

ways we change eurorack:

if you have financial stability - start designing modules because you want to, not to start a business. Sell cheaper. (could be double cost of production even, that's still often well below market rate). It also feels better because you're not tracking profit constantly, so you make what you want, and how much of it you want.

publicize costs/finances

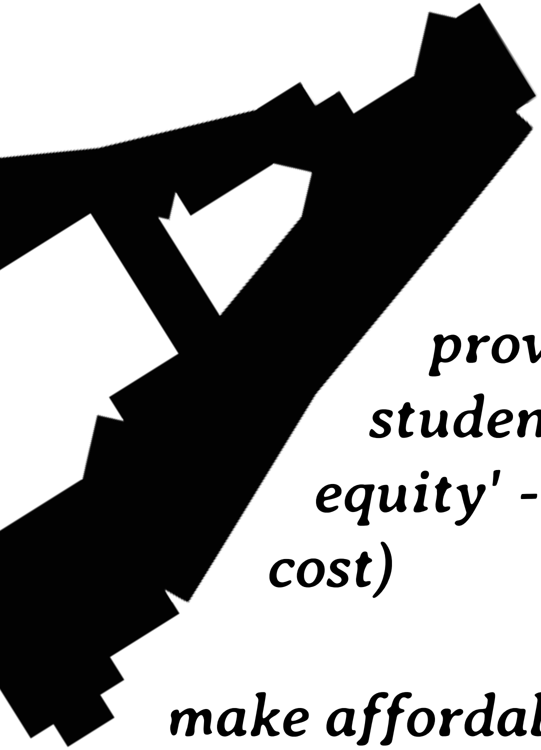
Transparency breeds accessibility. Most people don't really understand what a module actually is (a circuit board, panel, and components, designed and the then fabricated, usually in china). And how much it costs/why. Resistors cost 1/100 of a penny, jacks cost 50 cents, custom circuit boards cost 50 cents (depending) Supra's main processor (rp2350 which is cheap, new, and fastish) costs $1 - which, weirdly, is the same price as a single knob cap (?) (so you might be paying $5 in costs for knob caps on a synth and $1 for the processor) cheap faceplates cost $1 (circuit boards in disguise) expensive ones can be 20 times that. But, two years of support for someone to comfortable develop a module at 'market rate' in the US, ~$140,000, plus the cost for whoever assembles and distributes it. When you buy a module, you pay something for materials and assembly, but you're mostly paying for intellectual property - for technical skills, design, ideas. Knowing that can demonstrate to consumers that if they have design files (open source) and a little money or a group of people, they can have real affordability. Ordering circuit boards is 1000 times easier than designing a module - anyone can learn how to do it. If you can source the intellectual labor free (or make them yourself) bringing them into the physical world is cheaper and easier than you'd think.

(Note that intellectual labor is given for free all the time in the open source software world, and creative hardware can and should be the same, but we need to work on further normalizing and contributing to it.)

So, 2 expectations of makers

- publicize your costs
- when a module is discontinued, open source the design and manufacturing files. (ideally make a small post actually showing people how to order everything)

^neither of these cost money, and require pretty minimal effort



Some other things makers can do

provide sliding scale prices for modules, or at-cost for students, at cost for github contributors etc. (aka 'sweat equity' - if you contribute something to the project you pay at-cost)

make affordable cases - cases are a barrier to entry and cheap solutions are hard to come by - its a box that converts voltage with some crew holes, it doesn't need to cost much to make or sell
^similarly with power supply modules

make affordable audio/cv interfaces
- very key for people with limited eurorack gear. Very little is out there for this beyond Expert Sleepers - 4 channel audio/cv interface can be easily made for wayyy cheaper than you can find right now

Acknowledgments

Most of the acknowledgments here are the same as Pet Rock: *GalenDrew, AC Gaudette, Karp and Sam from Cutelab* **- all helped me with design and feedback, and this module would not be possible without them, for which I am incredibly thankful**

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SUPRA

SUPRA